Crossings never help: the Kobon conventions coincide, with certified values and the status of $K(14)$

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Abstract

The Kobon triangle problem asks for the maximum number of non-overlapping triangles determined by n straight lines. Three quantities appear in the literature under one name: $a_3^s(n)$, triangular faces of *simple* arrangements; $K(n)$, triangular faces of *arbitrary* arrangements (the object of Clément–Bader's definition, in which a counted triangle is "realized by 3 straight line segments"); and $K_{\text{gen}}(n)$, the maximum size of a family of triangles with pairwise disjoint interiors in which other lines *may* cross a counted triangle, so that a counted side may be subdivided. Trivially $a_3^s \le K \le K_{\text{gen}}$, and the classical segment-counting arguments bound only the first two.

Our main result is that the top two coincide: *crossings never help.* We prove a corner-descent lemma — every triangle bounded by three lines of an arrangement contains a triangular *face* of that arrangement — and deduce by a one-step exchange that $\text{face}(\mathcal{A}) = \text{broad}(\mathcal{A})$ for *every* arrangement, hence $K_{\text{gen}}(n) = K(n)$ for all n. The convention ambiguity is therefore harmless for values, and the non-simple segment bound of Felsner and Kriegel applies to interior-disjoint families with crossings. The collapse is pointwise false for *optima*: we exhibit arrangements, including an exact-rational 6-line arrangement attaining the record 7, whose maximum interior-disjoint family contains a crossed triangle.

The collapse also makes an all-degeneracy search sound: since crossings may be forbidden without loss of generality, a propositional encoding in which every parallel and multiple-point pattern is free decides the classical $K(n)$. We prove a capacity inequality valid with arbitrary parallels and multiple points,

$$3|S| + C + 2Q + \sum_p k_p(k_p - 4) \ \le\ n(n-2),$$

where C counts line/triangle interior incidences, together with an exact bounded-face budget; these drive the encoding. With them we certify $K(n)$ for every $n \le 10$ — $2, 5, 7, 11, 15, 21, 25$ from $n = 4$, with `drat-trim`-verified refutations of $K(n) + 1$ at $n = 4, \dots, 9$ — admitting every degeneracy. Three of these are new: at $n = 6, 8, 10$ the recorded value was a construction paired with a bound proved only for *simple* arrangements, while the bound valid for arbitrary arrangements is strictly larger (8, 16, 26); at $n = 10$ even the unpublished draft leaves 26 open, and at $n = 8$ the value 15 exceeds the exact simple maximum 14, so it is attained only non-simply. The same mismatch affects $n = 12$ and $n = 14$, where the constructions 38 and 54 exceed the exact simple maxima 37 and 53; we show the exactness claims recorded for them are unsupported. On the constructive side we verify, in exact rational arithmetic and with three implementations, a 14-line arrangement with 54 pairwise interior-disjoint triangles, and prove that every currently known extremal arrangement at $n = 11, 14, 18, 20$ attains its published count as its exact optimum. Finally we observe that the entry $a(14) = 54$ recorded as exact in OEIS A006066 on 21 August 2026 pairs a non-simple construction with a simple-arrangement upper bound; the rigorous status is $54 \le K(14) \le 55$, and that the same reading leaves $38 \le K(12) \le 39$, so $n = 12$ is in fact the

smallest open case. Each is closed by a single crossing-free instance with all degeneracies free. We also prove two equality-branch obstruction theorems ($n = 14$ and $n = 18$, machine-checked) and that the method cannot close the analogous branch at $n = 20$.

1 Three quantities

Let $\mathcal{A}$ be an arrangement of $n \geq 3$ distinct lines in the real affine plane, not all parallel (*essential*). A *segment* of $\mathcal{A}$ is a bounded maximal piece of a line between two consecutive crossing points.

Definition 1. (i) $a_3^s(n)$: the maximum number of triangular bounded faces of a *simple* arrangement of n lines (no two parallel, no three concurrent).

(ii) $K(n)$: the maximum, over all arrangements of n lines, of the number of triangles each of whose three sides is a single segment of $\mathcal{A}$, no two of them overlapping. Equivalently — since a line entering the interior of a triangle must cross two of its sides and thereby subdivide them — the maximum number of triangular bounded faces of an arbitrary arrangement.

(iii) $K_{\text{gen}}(n)$: the maximum, over all arrangements, of the size of a family S of triangles bounded by triples of lines of $\mathcal{A}$ whose open interiors are pairwise disjoint. Members of S may be crossed by other lines of $\mathcal{A}$.

Clearly $a_3^s(n) \leq K(n) \leq K_{\text{gen}}(n)$.

Clément and Bader [3] define a Kobon triangle as "a triangle that is realized by 3 straight lines segments and which does not overlap with other triangles", i.e. they work with K; their Lemma 1 argument enumerates the cases "side shared by two triangles" (which forces concurrency at an endpoint) and "segment used by no triangle", and their mod-6 improvement passes through *perfect configurations*, defined as arrangements of *pairwise intersecting* lines in which every segment is the side of exactly one triangle, and shown in their Lemma 2 to have no three concurrent lines. Their bound

$$K(n) \;\leq\; \lfloor n(n-2)/3 \rfloor - \mathbf{1}_{n \bmod 6 \in \{0,2\}} \tag{1}$$

is therefore a statement about K. Their case analysis does not treat a counted triangle whose sides are subdivided by crossings; by Theorem 4 it need not, since such triangles may be eliminated without loss. What remains is that (1) rests on an unpublished draft, and that its route — equality forces a perfect configuration, perfect configurations are simple and parallel-free (their Lemma 2), and the sharper simple bounds then contradict equality — has not been refereed. Bartholdi, Blanc and Loisel [1] and Blanc [2] bound a_3^s. Thus:

> *The base segment bound $\lfloor n(n-2)/3 \rfloor$ is stated for K including non-simple arrangements as Theorem 2(3) of Felsner and Kriegel [7] (so used, e.g., in [10]), with the proof deferred there to Roudneff [8]; the mod-6 improvement (1) for K rests on the unpublished draft [3]; the sharpest numerical bounds [1, 2] are proved for a_3^s only and hence do not apply to non-simple record arrangements. No bound had been proved for K_{gen}; Theorem 4 below shows none is needed, since $K_{\text{gen}} = K$.*

Remark 2 (provenance of the base bound)**.** For *simple* Euclidean arrangements the base bound is a two-line count: there are $n(n-2)$ bounded edges, and in a simple arrangement only one of the two cells along an edge can be a triangle (Grünbaum; equivalently Lemmas 5–6 of [7]), so $3p_3 \leq n(n-2)$. For arbitrary arrangements the situation is more delicate — an edge with a multiple endpoint *can* border two triangular cells — and [7] states the bound as Theorem 2(3) for every

Euclidean pseudoline arrangement while referring the proof to [8], whose theorem is the projective bound $p_3 \leq n(n-1)/3$ for arbitrary arrangements with $n \geq 9$ (false for $n \leq 8$). The reduction is not immediate in the direction needed: adjoining the line at infinity to a Euclidean arrangement of n lines and applying the projective bound gives only $p_3 \leq n(n+1)/3 - 3$, because Levi's theorem guarantees just three triangles along the added line whereas $n(n-2)/3$ requires n of them. We therefore cite Theorem 2(3) as the refereed statement and record that the numerical upper ends of Table 2 inherit that provenance — one more reason to prefer the self-contained certificates of §7, whose unsatisfiability verdicts bound K with every degeneracy free and cite nothing.

2 Crossings never help

Write face($\mathcal{A}$) for the number of triangular bounded faces of $\mathcal{A}$ (distinct faces have disjoint interiors, so this is also the largest interior-disjoint family of uncrossed triangles) and broad($\mathcal{A}$) for the largest family of triangles bounded by triples of lines of $\mathcal{A}$ with pairwise disjoint interiors, crossings permitted. Then $K(n) = \max_{\mathcal{A}} \operatorname{face}(\mathcal{A})$ and $K_{\text{gen}}(n) = \max_{\mathcal{A}} \operatorname{broad}(\mathcal{A})$, and face $\leq$ broad pointwise.

Lemma 3 (corner descent)**.** *Let T be a nondegenerate triangle bounded by three lines of $\mathcal{A}$. Then T contains a triangular face of $\mathcal{A}$.*

Proof. Induct on the number k of lines of $\mathcal{A}$ meeting $\operatorname{int} T$. If $k = 0$ then T is itself a face. Let $k \geq 1$ and let c meet $\operatorname{int} T$. The chord $c \cap T$ has its two endpoints on ∂T, and c contains at most one vertex of T: two vertices of T would lie on a common side line, forcing c to be that line. Two cases.

If c contains no vertex, its endpoints lie in the relative interiors of two distinct sides, on lines ℓ_a, ℓ_b say, and c cuts off the corner $\ell_a \cap \ell_b$: the corner piece is the triangle on $\{\ell_a, \ell_b, c\}$, nondegenerate because c misses that vertex.

If c contains the vertex $V = \ell_a \cap \ell_b$, its other endpoint lies in the relative interior of the opposite side, on ℓ_{opp}, and the two pieces are the triangles on $\{\ell_a, \ell_{\text{opp}}, c\}$ and $\{\ell_b, \ell_{\text{opp}}, c\}$; each is nondegenerate, since $c \cap \ell_a = V \notin \ell_{\text{opp}}$.

In both cases some piece $T' \subseteq T$ is a nondegenerate triangle bounded by three lines of $\mathcal{A}$, and every line meeting $\operatorname{int} T'$ meets $\operatorname{int} T$ while c does not meet $\operatorname{int} T'$; so the induction parameter of T' is at most $k-1$. Applying the inductive hypothesis to T' produces a triangular face inside $T' \subseteq T$. □

Theorem 4 (collapse)**.** face($\mathcal{A}$) = broad($\mathcal{A}$) *for every arrangement $\mathcal{A}$. Consequently $K_{\text{gen}}(n) = K(n)$ for every n.*

Proof. Let S be an interior-disjoint family of triangles of $\mathcal{A}$ with $|S| = \operatorname{broad}(\mathcal{A})$, and suppose $T \in S$ is crossed. By Lemma 3 there is a triangular face $F \subseteq T$; then $\operatorname{int} F \subseteq \operatorname{int} T$, so $S \setminus \{T\} \cup \{F\}$ is again interior-disjoint, has the same cardinality, and has one fewer crossed member. Iterating at most $|S|$ times yields an interior-disjoint family of uncrossed triangles, i.e. of triangular faces, of size broad($\mathcal{A}$). Hence face($\mathcal{A}$) $\geq$ broad($\mathcal{A}$), and the reverse inequality is trivial. Taking maxima over $\mathcal{A}$ gives $K_{\text{gen}}(n) = K(n)$. □

Corollary 5. $K_{\text{gen}}(n) \leq \lfloor n(n-2)/3 \rfloor$, *and* $K_{\text{gen}}(n) \leq \lfloor n(n-2)/3 \rfloor - \mathbf{1}_{n \bmod 6 \in \{0,2\}}$ *if* (1) *is granted. In particular the three quantities of Definition 1 satisfy $a_3^s \leq K = K_{\text{gen}}$.*

We could not locate either statement in the literature. The closest published relative is Lemma 4 of Leaños, Mbe Koua and Rivera [5], which reaches the same conclusion for *simple* arrangements under the extra hypothesis that one side of T is uncrossed, by a chord induction like the one above; the all-sides-crossed case is not covered there. The convention itself is flagged in only two places we found: Zamfirescu [6] defines *facial* triangles ($L \cap \operatorname{int} \Delta = \emptyset$ for all $L \in \mathcal{A}$) alongside general ones, and the Kobon variations recorded after Forge and Ramírez Alfonsín [4] ask that counted triangles not be crossed. Neither remarks that the count is unaffected. That the collapse is not automatic is shown by Zamfirescu's own problem: for *congruent* triangles the two conventions genuinely differ, her facial count being 14 at $n = 7$ against a larger general count [6]. The exchange of Theorem 4 uses interior-disjointness, which a congruence constraint destroys.

Two remarks delimit the result. First, the collapse is a statement about maxima, not about individual optima: an optimal family may genuinely contain a crossed triangle. For $n = 4$ this is typical rather than exceptional — in 296 of 300 random exact-rational generic arrangements some optimum of size $K(4) = 2$ uses a crossed triangle — and at $n = 6$ the arrangement

$$y = -\tfrac{9}{2}x - \tfrac{9}{2}, \quad y = -4x - 4, \quad y = -\tfrac{18}{5}x - 3, \quad y = -\tfrac{19}{6}x - \tfrac{19}{6}, \quad y = \tfrac{13}{3}x + 7, \quad y = 5x + \tfrac{4}{3}$$

(lines $0, 1, 3$ concurrent, no parallels) has face = broad = $7 = K(6)$ with the interior-disjoint optimum $\{012, 015, 024, 134, 135, 235, 245\}$ in which 012 is crossed by line 4; descent replaces 012 by the face 014 and recovers the face optimum, which is exactly the exchange of Theorem 4. Second, the descent is effective, and both statements have been machine-checked on exact-rational arrangements with planted parallel classes, triple points, quadruple points and pairs of triple points: over 872 arrangements at $n \le 7$ all 13,017 triangles descend to a face (depths up to 4) and no arrangement has face $<$ broad (`scratch/kobon/collapse_verify.py`).

Remark 6. Theorem 4 makes a search restriction sound: "some arrangement of n lines carries T pairwise interior-disjoint triangles" is equivalent to "some arrangement of n lines carries T triangles none of which is crossed". Forbidding crossings is therefore a WLOG, and it may be imposed on top of an encoding in which every parallel and multiple-point pattern remains free; §7 uses this.

This has an immediate consequence for the current record. On 21 August 2026 OEIS A006066 [14] was updated to list $a(14) = 54$ as exact, attributed to Maiorana's arrangement [12]. That arrangement is *not* simple: it has two triple points (on lines $\{0, 4, 10\}$ and $\{0, 11, 12\}$, sharing line 0) and no parallels. The matching upper bound 54 in the same table is the BBL value $\lfloor n(n - 7/3)/3 \rfloor$, proved for simple arrangements. For K, the peer-reviewed non-simple segment bound [7] gives 56 at $n = 14$, and the mod-6 improvement (1) of the draft [3] gives 55. Hence:

Proposition 7. *With the published literature,* $54 \le K(14) \le 56$ *unconditionally, and* $54 \le K(14) \le 55$ *if the draft [3] is granted; by Theorem 4 the same window holds for* $K_{\text{gen}}(14)$*. In every case* $a(14) = 54$ *is a lower bound paired with a simple-arrangement upper bound, so its exactness is not established.*

Contributions.

1. The collapse $K_{\text{gen}} = K$ (Theorem 4), from a corner-descent lemma, with the search WLOG it licenses (Remark 6) and exact-rational witnesses showing the corresponding pointwise statement is false.

2. A capacity theorem for K_{gen} (Theorem 12), with exact degeneracy penalty and crossing charge, and a closed-form ceiling (Theorem 16).

3. A certified initial segment with all degeneracies free: $K(4), \ldots, K(9) = 2, 5, 7, 11, 15, 21$, each with a `drat-trim`-verified refutation of $K(n) + 1$, plus $K(10) = 25$ from a prior cube cover (Theorem 19); of these $K(6), K(8), K(10)$ are first determinations (Corollary 9).

4. A hypothesis audit of the recorded small values (Proposition 8): at $n = 8, 12, 14$ the record exceeds the exact simple maximum, so the simple-only bound cited as "the upper bound" cannot apply.

5. An independently verified $K_{\text{gen}}(14) \geq 54$ (Theorem 20) and an exact optimality census of all known extremal arrangements (Theorem 22), which shows $K = K_{\text{gen}}$ on each of them.

6. Two equality-branch obstruction theorems with machine-checked finite certificates (Theorems 23, 24) and a proof that the same method cannot close the analogous branch at $n = 20$ (Proposition 25).

7. The status audit of $K(14)$ (Proposition 7) and the decision procedure that closes it.

3 Notation

For $\ell \in \mathcal{A}$ let q_ℓ be the number of lines parallel to ℓ and $Q = \frac{1}{2}\sum_\ell q_\ell$. A *multipoint* has multiplicity $k_p \geq 3$; N_k counts points of multiplicity k. On ℓ, r_ℓ is the number of distinct crossings, a_ℓ the number of multipoints, and $m_\ell = r_\ell - 1$ the number of bounded elementary gaps. For an admissible family S (Definition 1(iii)) set

$$\sigma_\ell = \#\{T \in S :\ \ell \text{ supports a side of } T\}, \qquad \gamma_\ell = \#\{T \in S :\ \ell \cap \operatorname{int}(T) \neq \emptyset\},$$

so $\sum_\ell \sigma_\ell = 3|S|$ and $C := \sum_\ell \gamma_\ell$ is the total number of line/triangle interior incidences. Write $T = |S|$ and $\Delta = n(n-2) - 3T$.

4 What is actually proved for small n

The recorded values are widely treated as exact. They are not, and the reason is a hypothesis mismatch that Theorem 4 lets us state cleanly, since only one convention now exists.

Write $\bar{a}_3^s(n)$ for the exact maximum over *simple* affine arrangements. Blanc [2] (Theorem 1 for the bound, Theorem 3 for attainment, computer-assisted) and [1] (Theorem 1.4) give $\bar{a}_3^s(8) = 14$, $\bar{a}_3^s(11) = 32$, $\bar{a}_3^s(12) = 37$ and $\bar{a}_3^s(14) = 53$. Meanwhile the even-n bound $\lfloor n(n - 7/3)/3 \rfloor$ of [1, Thm. 1.1] — proved under the explicit hypothesis "our sole interest is with simple arrangements, i.e. arrangements without multiple intersections" — takes the values $7, 15, 25, 38, 54$ at $n = 6, 8, 10, 12, 14$, which are exactly the recorded constructions.

Proposition 8. *For $n \in \{8, 12, 14\}$ every arrangement attaining the recorded value (15, 38, 54 respectively) is non-simple, because that value exceeds $\bar{a}_3^s(n) \in \{14, 37, 53\}$. Consequently the simple-arrangement bound $\lfloor n(n-7/3)/3 \rfloor$ of [1, Thm. 1.1], which coincides with the recorded value at those n, cannot be applied to the arrangements in question, and the exactness claims that invoke it — the upper-bound column of OEIS A006066 [14] at $n = 12$ and its comment of 13 August 2026 at $n = 14$, and the corresponding table in MathWorld [13] — are unsupported. The published bounds valid for arbitrary arrangements are $\lfloor n(n-2)/3 \rfloor = 16, 40, 56$ [7], and $15, 39, 55$ if the unpublished draft [3] is granted.*

Proof. A simple arrangement has at most $\bar{a}_3^s(n)$ triangular faces by definition of $\bar{a}_3^s$, and $15 > 14$, $38 > 37$, $54 > 53$. The remaining statements are the cited hypotheses and values. □

At $n = 6$ and $n = 10$ the construction equals $\bar{a}_3^s(n)$, so it may be simple; but the same gap remains on the upper side, since $\lfloor n(n-2)/3 \rfloor$ is 8 and 26 there. Collecting:

Corollary 9. *Before the certificates of §7, the values determined by the refereed literature were* $K(n)$ *for* $n \le 5$ *and* $n \in \{7, 9, 13\}$*, where* $\lfloor n(n-2)/3 \rfloor$ *is attained, together with* $K(11) = 32$ *[9]. For the even cases* $n \in \{6, 8, 10, 12, 14\}$ *the record was a construction paired with a bound proved only for simple arrangements. Granting the unpublished draft [3] would settle* $n = 6$ *and* $n = 8$*, since* (1) *gives* 7 *and* 15 *there; it would not settle* $n = 10, 12, 14$*, where* (1) *gives* $26, 39, 55$ *against constructions* $25, 38, 54$.

5 The capacity theorem

Lemma 10 (incidence identity). $m_\ell + 2a_\ell = n - 2 - q_\ell - \sum_{p \ni \ell}(k_p - 4)$ *for every* $\ell \in \mathcal{A}$.

Proof. The $n - 1 - q_\ell$ lines meeting ℓ do so in r_ℓ distinct points, and a multipoint p on ℓ absorbs $k_p - 1$ of them into one point, so $n - 1 - q_\ell = r_\ell + \sum_{p \ni \ell}(k_p - 2)$. Hence $m_\ell + 2a_\ell = r_\ell - 1 + 2a_\ell = n - 2 - q_\ell - \sum_{p \ni \ell}(k_p - 2) + 2a_\ell$, and a_ℓ is the number of summands. □

Lemma 11 (per-line capacity). $\sigma_\ell + \gamma_\ell \le m_\ell + 2a_\ell$ *for every* $\ell \in \mathcal{A}$ *and every admissible* S.

Proof sketch. Each $T \in S$ with a side on ℓ determines the side interval $I_T \subset \ell$; each T crossed by ℓ determines the chord interval $J_T = \ell \cap \operatorname{int}(T)$. Disjointness of interiors makes the J_T pairwise disjoint, makes each J_T disjoint from the interiors of side intervals on the same side of ℓ, and makes side intervals on a fixed side of ℓ interior-disjoint. Every such interval contains at least one elementary gap, and two intervals can claim the same gap only when they lie on opposite sides of ℓ and abut a common multipoint of ℓ, each of which admits at most two coincidences. Charging one gap per interval and two tokens per multipoint yields the claim; the exhaustive case analysis is carried out in the companion document [15, §3] and was re-derived by an adversarial audit over 3,016 arrangements and 812,486 admissible families with 5,900,863 exact linewise checks. □

Theorem 12 (capacity). *For every essential arrangement of* n *distinct affine lines and every admissible family* S*,*

$$3|S| + C + 2Q + \sum_p k_p(k_p - 4) \;\le\; n(n-2). \tag{2}$$

Proof. Sum Lemma 11 over all lines and substitute Lemma 10; the left side gives $3|S| + C$, and on the right $\sum_\ell q_\ell = 2Q$ while each multipoint lies on exactly k_p lines, so $\sum_\ell \sum_{p \ni \ell}(k_p - 4) = \sum_p k_p(k_p - 4)$. □

Setting $C = 0$ recovers, with explicit degeneracy corrections, the classical segment bound for K; the point of (2) is that the crossing charge makes it valid for K_{gen}.

Remark 13. Neither correction term can be replaced by the naive alternatives: the variants with $\binom{k_p - 2}{2}$ or $(k_p - 2)^2$ in place of $k_p(k_p - 4)$ are false, with exact counterexamples in [15, §6]. The all-parallel class is the unique inessential exception.

Corollary 14. *If* $\mathcal{A}$ *has no triple point then* $|S| \le \frac{1}{3}\big(n(n-2) - C - 2Q - \sum_{k \ge 4} N_k k(k-4)\big) \le n(n-2)/3$*. In general* $|S| \le n(n-2)/3 + N_3 - \frac{1}{3}\big(C + 2Q + \sum_{k \ge 5} N_k k(k-4)\big)$*, so triple points are the only degeneracy that can raise the ceiling, by at most one triangle each.*

6 A closed-form ceiling for K_{gen}

Lemma 15 (face budget)**.** $|S| + C + Q + \sum_p \binom{k_p-1}{2} \le \binom{n-1}{2}$.

Proof. $\mathcal{A}$ has exactly $\binom{n-1}{2} - Q - \sum_p \binom{k_p-1}{2}$ bounded faces. The interior of $T \in S$ contains at least $1 + c_T$ bounded faces, where c_T is the number of lines crossing it, and distinct members have disjoint interiors, so they consume disjoint face sets; summing gives $|S| + C$. □

Theorem 16 (closed-form ceiling)**.** $K_{\text{gen}}(n) \le \lfloor (n-2)(5n-3)/12 \rfloor$ *for all* $n \ge 3$. *In the combined inequality used, multiplicity* 3 *has coefficient exactly* 0 *and every* $k \ge 4$ *has coefficient* $k(k-4) + 3\binom{k-1}{2} > 0$.

Remark 17. Since $5n - 3 > 4n$ for $n > 3$, this ceiling is weaker than $\lfloor n(n-2)/3 \rfloor$ at every $n \ge 4$, so Corollary 5 supersedes it numerically. We retain it because it is proved directly from the capacity inequality and the face budget without the exchange argument, and because those two inequalities — not the closed form — are what the encoding of §7 propagates.

Proof. Add (2) to three times Lemma 15:

$$6|S| + 4C + 5Q + \sum_{k\ge3} N_k\Big(k(k-4) + 3\binom{k-1}{2}\Big) \le n(n-2) + 3\binom{n-1}{2} = \frac{(n-2)(5n-3)}{2}.$$

The bracket is $0, 9, 23, 42, \dots$ for $k = 3, 4, 5, 6$, hence nonnegative. □

Lemma 18 (reduction at zero slack)**.** *If* $\Delta = 0$ *and* $\mathcal{A}$ *has no triple point, then* $C = 0$, $Q = 0$, *every multipoint has* $k_p = 4$, *and every member of* S *is a bounded triangular face; in particular* $|S| \le K(n)$ *is attained by faces.*

Proof. (2) forces $C + 2Q + \sum_{k\ge4} N_k k(k-4) \le 0$ with all terms nonnegative. □

Among known values, (2) is attained exactly at $n = 3, 5, 9, 15, 17$ — precisely the n where the Tamura value $n(n-2)/3$ is realised — and there all three quantities of Definition 1 coincide.

7 Certified small values, all degeneracies free

The decision problem "is there an n-line arrangement with an admissible family of size T?" is encoded in propositional logic over slope-sorted combinatorial data: variables for crossing versus parallel, concurrency of triples, the order of crossings along each line, the selection, and triangle/line crossing incidences, with the per-line inequalities of Lemma 11 and the budget of Lemma 15 as exact cardinality constraints. All degeneracies are free, so a verdict covers crossed and degenerate configurations simultaneously.

Theorem 19. *In the convention of Definition 1(iii),*

$$K_{\text{gen}}(4), \dots, K_{\text{gen}}(9) \;=\; 2,\ 5,\ 7,\ 11,\ 15,\ 21,$$

and each value is certified: the instance $(n, K(n)+1)$ *is unsatisfiable with a* `drat-trim`*-verified proof (Table 1), and the matching lower bounds are the classical constructions. Together with the prior eleven-cube cover for* $n = 10$ *this gives* $K(n)$ *for all* $n \le 10$*; the* $n = 10$ *value is confirmed a second time, through a different encoding, by a single monolithic refutation of* $(10, 26)$ *in the crossing-free*

n	$K_{\text{gen}}(n)$	refuted T	variables / clauses	core clauses	core lemmas	verify (s)
4	2	3	207 / 994	90	131	0.05
5	5	6	972 / 5,257	424	558	0.04
6	7	8	3,655 / 20,681	4,541	7,018	0.48
7	11	12	10,682 / 63,405	12,255	31,595	4.47
8	15	16	27,451 / 165,367	31,218	411,278	117.6
9	21	22	61,569 / 377,405	66,346	2,150,510	1060.3

Table 1: Certified initial segment. Every row is a `s VERIFIED` `drat-trim` run on a kissat proof of the refutation of $T = K_{\text{gen}}(n) + 1$; proofs range from 12 kB to 328.5 MB and the resolution counts from 582 to $8.7 \cdot 10^7$. Ledger: `scratch/kobon/ladder_certificates.json`.

*formulation of Theorem 4 (*131 314 *variables,* 792 462 *clauses, unsatisfiable in* 31 *minutes, discovery-level: the DRAT certificate for this value is the cube cover). By Corollary 9 the rungs at* $n = 6$*,* $n = 8$ *and* $n = 10$ *are the first determinations of those values from refereed inputs: they rule out* 8*,* 16 *and* 26 *triangles respectively, which no simple-arrangement bound can exclude. At* $n = 10$ *this is the first determination outright, since even* (1) *leaves* 26 *open; at* $n = 6, 8$ *it removes the dependence on the unpublished draft. And* $K(8) = 15$ *is attained only non-simply (Proposition 8), so the certified statement lives entirely in the degenerate regime.*

These statements are strictly stronger than the corresponding classical ones, since they also exclude crossed families: e.g. $K_{\text{gen}}(8) = 15$ rules out the value $16 = \lfloor 8 \cdot 6/3 \rfloor$ left open for K_{gen} by the segment bound, and $K_{\text{gen}}(6) = 7$ rules out 8.

8 A verified lower certificate at $n = 14$

Theorem 20. $K_{\text{gen}}(14) \geq 54$*, witnessed by an arrangement with exact rational coefficients.*

Proof. From the SHA-256-pinned source file of [12] we verified: the 14 lines are pairwise non-proportional; the computed multipoint census is exactly two triple points sharing a line, with $Q = 0$, matching the file's declared data; exactly 54 triples have interiors met by no other line, computed with two independent predicates, the second of which also rejects the case of a line through a vertex separating the opposite side; and the resulting family passes the audited exact-rational checker `verify_selection`, which validates non-degeneracy and pairwise interior-disjointness in homogeneous rational coordinates. The same verification succeeds for all fifteen published solutions. □

9 Crossing incidences are exactly reified

The experiments that force or forbid crossings rest on a single literal being exact, so we record what it encodes. For a triple t of lines and a line $r \notin t$, let $\mathrm{TC}(t, r)$ be the variable defined by

$$\mathrm{TC}(t, r) \iff S(t) \wedge \Big(\bigvee_{e \subset t} \mathrm{SA}(e, r)\Big) \wedge \Big(\bigvee_{e \subset t} \mathrm{SB}(e, r)\Big), \tag{3}$$

where $S(t)$ says t is in the selected family and $\mathrm{SA}(e, r)$, $\mathrm{SB}(e, r)$ say that the vertex of t opposite[1] lies strictly above, respectively strictly below, r. Both sidedness literals are exact in the model.

[1] Indexed by the pair e of lines meeting there.

Lemma 21. *In every arrangement,* $\mathrm{TC}(t, r)$ *holds precisely when* t *is selected and* r *meets the open interior of* t*.*

Proof. If some vertex of t is strictly above r and another strictly below, then r crosses the two sides joining them at relative interior points, and the open chord between those points lies in the open interior. Conversely, suppose r meets the open interior. Then r cannot contain two vertices of t: two vertices span a side, which lies on a line of t, and $r \notin t$. So at most one vertex is on r, and the chord separates the remaining two strictly, giving one strictly above and one strictly below. The conjunct $S(t)$ is asserted by the first clause of (3). □

The point of the lemma is that TC is attached to a *selected* triangle: a disjunction over TC literals asserts that the family exhibited by the model has a crossed member, not merely that the arrangement crosses some candidate triple. Cevians and lines through a vertex are handled by the "at most one vertex on r" case rather than excluded.

10 Exact optimality of the known witnesses

For a *fixed* arrangement, the largest admissible family is the independence number of the overlap graph on non-degenerate triples, with edges joining pairs of intersecting interiors. We compute it exactly: interiors are compared in homogeneous rational arithmetic, and optimality is certified by SAT (a family of size K is a model, UNSAT at $K + 1$ certifies the optimum).

Theorem 22. *For each of the following the exact admissible optimum equals the published count, with certified optimality: the fifteen* 14*-line* 54*-triangle arrangements of [12]; two exact rational* 53*-triangle* 14*-line reconstructions; an* 11*-line* 32*-triangle arrangement; an* 18*-line* 93*-triangle reconstruction; and a* 20*-line* 116*-triangle reconstruction. In particular* $K = K_{\text{gen}}$ *on every known extremal arrangement.*

By Lemma 21 we may also ask whether an *optimal* family must be crossing-free, by conjoining $\bigvee_{t,r} \mathrm{TC}(t, r)$ with exact selection of $K(n)$ triangles. This is unsatisfiable for $n \in \{5, 7, 8, 9\}$ and satisfiable for $n \in \{4, 6\}$, so crossed optima exist but only at the smallest sizes; at $n = 4$ they are generic, occurring in 296 of 300 random exact-rational arrangements, and at $n = 6$ an explicit arrangement carries both a crossed and an uncrossed optimal family of size 7. The satisfiable cases are therefore not artefacts of degeneracy, and by Theorem 4 they never increase the optimum.

Consequently any improvement on the records must come from new arrangements, and by Corollary 14 any improvement past the Tamura value must use triple points — which is exactly the mechanism of Maiorana's step from 53 to 54.

11 Equality-branch obstructions

When Δ is small the admissible degeneracy patterns form a short list, and saturation of Lemma 11 forces the multiplicities m_i of double-extreme vertices at consecutive direction classes to satisfy

$$m_{i-1} + m_i = 2s_i, \qquad 0 \le m_i \le s_i s_{i+1}, \qquad m_i \in \mathbb{Z}, \tag{4}$$

cyclically in the class sizes $(s_1, \dots, s_R)$. For odd R the rational solution is unique; for even R solvability requires $\sum_i (-1)^i 2s_i = 0$.

Theorem 23 ($n = 14$)**.** *No arrangement of* 14 *lines with exactly three parallel pairs, or one parallel triad, and with no three lines concurrent admits an admissible family of* 54 *triangles.*

Proof. Paired layout: sizes are three 2's and eight 1's, $R = 11$ odd, and the unique solution of (4) violates an interval constraint for each of the $\binom{11}{3} = 165$ placements. Triad layout: one 3 and eleven 1's, $R = 12$ even, and the alternating sum is $\pm 4 \neq 0$ at each of the 12 placements. Both enumerations are certified by an exact integer program (pure standard library, no floating point). $\square$

Theorem 24 ($n = 18$)**.** *No arrangement of* 18 *lines with exactly three parallel pairs, or one parallel triad, and with no three lines concurrent admits an admissible family of* 94 *triangles.*

Proof. Same mechanism with $R = 15$ and $R = 16$: all $\binom{15}{3} = 455$ paired placements violate an interval constraint (gap-parity classes $140 + 90 + 225$, 19 dihedral orbits with exact multiplicities, reflection quotients 231 and 9 all asserted by the certificate), and all 16 triad placements have alternating sum ± 4. $\square$

Proposition 25 (the method stops at $n = 20$)**.** *At $n = 20$, $T = 117$ we have $\Delta = 9$ and $2Q = 6$, leaving three units of slack, so* (4) *must be relaxed by a defect vector, and feasible defects exist: for the paired layout ($R = 17$) a defect of* 2 *at one doubled class with run lengths $(u, 0, 14 - u)$, e.g.* $(6, 0, 8)$*; for the triad layout ($R = 18$) a defect of* 4 *with all $m_i = 1$. Hence no argument using only* (4) *can exclude $T = 117$ at $n = 20$.*

12 The two open small cases, and how to close them

Table 2 collects what is proved. Theorem 4 simplifies the picture decisively: there is only one quantity above a_3^s, so one instance per n suffices, and the rungs above it that a bound for K_{gen} alone would leave open do not exist.

Reading the table with the simple-arrangement entries discarded leaves exactly two open cases below $n = 18$. At $n = 11$ the value is settled [9]; at $n = 13$ the peer-reviewed bound $\lfloor 13 \cdot 11/3 \rfloor = 47$ equals the construction, so $K(13) = 47$. But at $n = 12$ the construction gives 38 while the bounds valid for K give 40 [7] and 39 (1); the OEIS upper bound 38 is a simple-arrangement value. So $K(12) \in \{38, 39\}$, and $n = 12$ — not $n = 14$ — is the smallest genuinely open case. At $n = 14$ the same reading gives $K(14) \in \{54, 55\}$.

Each case is one decision instance, and by Remark 6 it may forbid crossings while leaving every parallel and multiple-point pattern free:

- $(12, 39)$: unsatisfiability gives $K(12) \leq 38$, hence $K(12) = 38$ with the construction. This is self-contained — it does not use (1) or [7] — so it also removes the reliance of the $n = 12$ row on an unpublished draft.
- $(14, 55)$: unsatisfiability gives $K(14) = 54$, making the OEIS entry a theorem; satisfiability gives, after straightening, a 55-triangle record. The crossing-free form has 1,349,454 variables and 7,829,001 clauses.

Both are running at the time of writing, together with the crossing-free $(11, 33)$ as an independent check against the known value.

13 Open problems

1. **Decide** $(12, 39)$ **and** $(14, 55)$, closing the two open small cases. By Theorem 4 and Remark 6 each is one crossing-free instance with all degeneracies free, and no further rungs exist above it. Unsatisfiability at $(12, 39)$ proves $K(12) = 38$ without (1); unsatisfiability at $(14, 55)$ proves $K(14) = 54$.

n	constr.	$\bar{a}_3^s$	(1)	FK	status
6	7	7	7	8	$K(6) = 7$ certified here (DRAT)
8	15	14	15	16	$K(8) = 15$ certified here (DRAT)
9	21	21	21	21	$K(9) = 21$; FK attained
10	25	25	26	26	$K(10) = 25$ certified here (cube cover)
11	32	32	33	33	$K(11) = 32$ [9]
12	**38**	37	39	40	$[38, 40]$; $(12, 39)$ running
13	47	47	47	47	$K(13) = 47$; FK attained
14	**54** (verified)	53	55	56	$[54, 56]$; $(14, 55)$ running
18	93	93	95	96	$[93, 96]$
20	116	116	119	120	$[116, 120]$

Table 2: Constructions are lower bounds for $K = K_{\text{gen}}$ (Theorem 4). Column $\bar{a}_3^s$ is the exact maximum over *simple* arrangements [1, 2]: where the construction exceeds it (bold: $n = 8, 12, 14$) the record is attained only non-simply, so the simple-only bound $\lfloor n(n - 7/3)/3 \rfloor$ — which equals the construction at every even n here and is what the OEIS upper-bound column reports — cannot apply (Proposition 8). The bounds that do apply are (1), resting on the unpublished draft [3], and FK $= \lfloor n(n-2)/3 \rfloor$ [7] (see Remark 2). Rows marked *certified here* are settled by unsatisfiability with every degeneracy free, citing nothing.

2. **A mod-6 refinement for arbitrary arrangements**, i.e. derive the -1 correction of (1) inside the framework of (2), removing the reliance on an unpublished draft. This is now the only gap between the peer-reviewed bound $\lfloor n(n-2)/3 \rfloor$ [7] and the numbers used in Table 2, and it is exactly what decides $n \equiv 0, 2 \pmod 6$ — including $n = 14$, where it is the difference between the windows $[54, 55]$ and $[54, 56]$.

3. **Defect-tolerant obstructions.** Proposition 25 localises the missing ingredient: a constraint on where a defect may sit.

4. **Sharpen the capacity inequality with degeneracies.** Equality in (2) occurs only at $N_3 = 0$ throughout the known census, yet the inequality permits $\sum_p k_p(k_p - 4) < 0$. A structural proof that the $2a_\ell$ tokens of Lemma 11 are never simultaneously realisable would recover $\lfloor n(n-2)/3 \rfloor$ from the capacity machinery alone, i.e. without the exchange argument of Theorem 4 and without [7].

5. **Pointwise structure.** Theorem 4 equates the two maxima but not the optima: crossed optima exist (§3). Which arrangements admit *only* crossed optima of maximum size, and can the descent be made canonical?

A Artefacts and replay

Release bundle `math/kobon/release/kobon-2026-08/` with `MANIFEST.sha256`; documentation hub `math/kobon/docs/`.

- `verification/maiorana_verify.py` — fail-closed certification of Theorem 20 (exit-0 transcript included).
- `verification/sweep_all_verify.py` — all fifteen witnesses.

- `verification/max_packing.py` — exact overlap graph and SAT optimality, used for Theorem 22.
- `scratch/kobon/broad_ladder.py` — ladder prober used for Theorem 19; instances `ladder_n8_t16.cnf`, `ladder_n9_t22.cnf` and the DRAT proofs.
- `proofs/appendix_independent_check.py`, `proofs/appendix_check_n18.py` — exact integer certificates for Theorems 23 and 24.
- `proofs/defect_lemma.md` — analysis behind Proposition 25.
- `scripts/engine.py` — exact rational geometry core and CNF builder.

References

[1] N. Bartholdi, J. Blanc, S. Loisel, *On simple arrangements of lines and pseudo-lines in $\mathbb{P}^2$ and $\mathbb{R}^2$ with the maximum number of triangles*, Contemp. Math. **453** (2008), 105–116; arXiv:0706.0723.

[2] J. Blanc, *The best polynomial bounds for the number of triangles in a simple arrangement of n pseudo-lines*, Geombinatorics **21** (2011), no. 1, 5–14; preprint arXiv:0801.2845.

[3] G. Clément, J. Bader, *Tighter upper bound for the number of Kobon triangles*, draft, ETH Zürich, 21 December 2007.

[4] D. Forge, J. L. Ramírez Alfonsín, *Straight line arrangements in the real projective plane*, Discrete Comput. Geom. **20** (1998), no. 2, 155–161.

[5] J. Leaños, M. K. Ndjatchi, L. M. Rivera-Martínez, *Simple Euclidean arrangements with one (≥ 5)-gon*, arXiv:1008.4598 (2010); Lemma 4.

[6] C. T. Zamfirescu, *Congruent triangles in arrangements of lines*, Ars Math. Contemp. **14** (2018), no. 2, 359–373.

[7] S. Felsner, K. Kriegel, *Triangles in Euclidean arrangements*, Discrete Comput. Geom. **22** (1999), no. 3, 429–438.

[8] J.-P. Roudneff, *The maximum number of triangles in arrangements of pseudolines*, J. Combin. Theory Ser. B **66** (1996), no. 1, 44–74.

[9] P. Savchuk, *Constructing optimal Kobon triangle arrangements via table encoding, SAT solving, and heuristic straightening*, arXiv:2507.07951 (2025).

[10] R. Parpalak, D. Utkin, *Enumeration and classification of triangle-maximal pseudoline arrangements*, arXiv:2607.29236 (2026).

[11] R. Parpalak, D. Utkin, *The $18 \cdot 2^t + 1$ triangle-maximal series of straight lines*, arXiv:2604.22035 (2026).

[12] A. Maiorana, *Kobon triangles, $k = 14$*, dataset, `https://github.com/rufio72/kobon_triangles_k14`, commit `e47c7cfd`.

[13] E. W. Weisstein, *Kobon Triangle*, MathWorld—A Wolfram Web Resource, `https://mathworld.wolfram.com/KobonTriangle.html`, accessed 22 August 2026.

[14] OEIS Foundation Inc., *Sequence A006066: Kobon triangles*, `https://oeis.org/A006066` (revision #227, 21 August 2026).

[15] Companion technical document `math/kobon/paper_capacity.md`: full case analysis of Lemma 11, audit protocols, refutations of weaker penalty forms, and the complete experiment ledger.